ANSWER KEY & MARKING SCHEME SET 1

Chapter 12: **Magnetic Effects of Electric Current** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Both A and R are true and R is the correct explanation of A.	At a crossing there would be two directions of the field.
2	**(c)** 90 degrees	Perpendicular is the maximum.
3	**(d)** 100 Hz	Proportional to speed.
4	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true; the deflection is evidence of the field, not its explanation.
5	**(d)** zero	No change of linked field.
6	**(d)** 3 E	Proportional to rate of change.
7	**(a)** Both A and R are true and R is the correct explanation of A.	Needed for a temporary magnet.
8	**(b)** develops an induced current while the magnet moves	A changing field links the coil only during the motion.
9	**(a)** Both A and R are true and R is the correct explanation of A.	If two lines crossed, there would be two directions of the field at the point of intersection.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true; the field pattern outside is like that of a bar magnet, which is not explained by the uniform field inside.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Force is perpendicular to the velocity (1); No work is done, so the speed is constant (1).	1+1
12	I = 2200 / 220 = 10 A (1); The 15 A circuit is suitable (1).	1+1
13	It disconnects the appliance from the high potential (1); The appliance stays safe when switched off (1).	1+1
14	T = 1 / 60 = 0.0167 s (1); Reversals = 120 per second (vs 100) (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Force arises from the component of the field perpendicular to the wire (1); It is maximum when the wire is perpendicular (1); It is zero when the wire is parallel (1).	1+1+1
16	Metal case can become live due to a fault (1); The earth wire is a low-resistance path (1); The fuse blows and the user is safe (1).	1+1+1
17	Current produces a magnetic field (Oersted) (1); Changing magnetic field produces current (Faraday) (1); Force on a conductor in a field links the two (1).	1+1+1
	OR: Inside: field lines are nearly parallel and in one direction (1); Outside: lines go back in the opposite direction (1); Each element of the wire adds to the field (1).	1+1+1
18	Straight wire: concentric closed circles around the wire (1); Bar magnet: lines emerge from N and enter S (1); Both have closed loop fields (1).	1+1+1
	OR: Increase the number of turns (1); Use a soft iron core (1); Use a thicker core or a larger current (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Hans Christian Oersted (1). (ii) An electric current produces a magnetic field around the wire (1). (iii) Right-hand thumb rule (1). (iv) Anticlockwise (concentric circles) as seen from above (1); the field strength decreases as the distance from the wire increases (and increases with current) (1).	1+1+1+2
20	A rectangular coil is placed between the poles of a magnet (1); Current in the coil gives forces on its two arms in opposite directions (1); The coil rotates (Fleming's left-hand rule) (1); Split rings reverse the current every half rotation (1); So the coil continues to rotate in the same direction (1).	1+1+1+1+1

	OR: Deflection increases with speed (1); It is proportional to the speed here (2 : 4 : 6 = 1 : 2 : 3) (1); The induced emf depends on the rate of change of the field (1); With double the turns the emf doubles (1); Deflections would be about 4, 8 and 12 divisions (1).	1+1+1+1+1

Mistakes to avoid:

- Mixing up Fleming's left-hand rule (force on a current-carrying conductor) and right-hand rule (induced current in a moving conductor).
- Saying that the field lines inside a magnet go from N to S; outside they go N to S, but inside they go from S to N (closed loops).
- Believing that a stationary magnet inside a coil induces a current; induction requires a **change** of magnetic field.
- Writing the frequency of Indian AC as 60 Hz or 100 Hz; it is 50 Hz, and the direction reverses every $\frac{1}{100}$ s.
- Connecting the fuse in the neutral or earth wire; the fuse is placed in the live wire.
- Confusing the colours: live is red, neutral is black, and earth wire is green; confusing short circuit with overloading.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 12: **Magnetic Effects of Electric Current** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** clockwise	Clockwise appears as a south pole.
2	**(d)** 6000	100 per s x 60.
3	**(a)** Both A and R are true and R is the correct explanation of A.	Nothing is cut.
4	**(b)** anticlockwise when seen from above	Fingers curl anticlockwise from above.
5	**(d)** blows because I = 6.8 A	I = 1500 / 220 = 6.8 A > 5 A.
6	**(a)** Both A and R are true and R is the correct explanation of A.	Lower loss explains the preference.
7	**(a)** zero	No force when the motion is parallel to the field.
8	**(d)** A and B together	A and B together draw 1.5 kW, which is more than 1100 W.
9	**(c)** A is true but R is false.	India's AC has frequency 50 Hz (reverses direction every $\frac{1}{100}$ s); R is false.
10	**(d)** A is false but R is true.	The fuse is connected in the live wire; A is false, R is true.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Induced emf is larger (1); More turns link more of the changing field (1).	1+1
12	It reverses the direction of current in the coil every half rotation (1); So the coil keeps rotating in the same direction (1).	1+1
13	Bring a compass near the wire (1); Deflection of the needle shows a current (1).	1+1
14	Iron is a magnetic material (1); Copper is not magnetised by an external magnet (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Total power = 3 kW (1); I = 3000 / 220 = 13.6 A (1); More than 5 A, so the fuse blows (1).	1+1+1
16	Bring the needle near an end (1); If the north pole of the needle is attracted, the end is a south pole (1); Reversing the current reverses the polarity (1).	1+1+1
17	The two sides of the coil exchange positions every half rotation (1); The direction of the induced current reverses each half rotation (1); Hence the output is alternating (1).	1+1+1
	OR: Move the magnet slowly: small deflection (1); Move it quickly: larger deflection (1); The rate of change of the field decides the emf (1).	1+1+1
18	Charge moves perpendicular to the field (1); Force is zero when it moves along the field (1); Maximum at 90 degrees (1).	1+1+1
	OR: The induced emf doubles (1); The frequency is unchanged (1); More turns link more field lines (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Electromagnetic induction (1). (ii) Michael Faraday (1). (iii) Current is induced only when the magnetic field through the coil changes, which does not happen when the magnet is at rest (1). (iv) Move the magnet faster, use a stronger magnet, or increase the number of turns of the coil (any two) (1+1).	1+1+1+2
20	(a) Place a compass near the solenoid; switch on the current (1); Needle deflects, so current produces a magnetic field (1); (b) Connect the solenoid to the galvanometer (1); Move the magnet in and out (1); Deflection shows induced current from a changing field (1).	1+1+1+1+1
	OR: (a) Metal body may stay live, causing a shock (1); (b) Live wire remains connected when the fuse melts (1); (b) The appliance remains at high potential (1); (c) Appliance stays live even when switched off (1); Hence the fuse and switch must be in the live wire (1).	1+1+1+1+1

Mistakes to avoid:

- Mixing up Fleming's left-hand rule (force on a current-carrying conductor) and right-hand rule (induced current in a moving conductor).
- Saying that the field lines inside a magnet go from N to S; outside they go N to S, but inside they go from S to N (closed loops).
- Believing that a stationary magnet inside a coil induces a current; induction requires a **change** of magnetic field.
- Writing the frequency of Indian AC as 60 Hz or 100 Hz; it is 50 Hz, and the direction reverses every $\frac{1}{100}$ s.
- Connecting the fuse in the neutral or earth wire; the fuse is placed in the live wire.
- Confusing the colours: live is red, neutral is black, and earth wire is green; confusing short circuit with overloading.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 3

Chapter 12: **Magnetic Effects of Electric Current** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** At high potential with respect to earth	220 V.
2	**(a)** Both A and R are true and R is the correct explanation of A.	Breaking the live wire isolates the appliance.
3	**(a)** A stationary charge	Only moving charges and magnets.
4	**(a)** the fuse blows and the person is safe	Large current passes to earth.
5	**(a)** Both A and R are true and R is the correct explanation of A.	Force perpendicular to motion does no work.
6	**(b)** loses most of the magnetism if it is soft iron	Soft iron.
7	**(c)** decreases	Core enhances the field.
8	**(a)** Both A and R are true and R is the correct explanation of A.	R explains why the commutator is needed.
9	**(a)** Both A and R are true and R is the correct explanation of A.	The leakage current flows to the earth instead of through the user's body.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	Both true; R is a further fact about the direction of the force, not a reason for A.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Field is proportional to the number of turns per unit length (1); B has double the field (1).	1+1
12	It acts like a bar magnet (1); It aligns with the earth's field (1).	1+1
13	Field is strongest near the poles (1); Field lines are closer there (1).	1+1
14	Each piece has its own north and south poles (1); Isolated poles are not possible (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	One cycle for one rotation (1); Frequency = 50 Hz (1); Period = 0.02 s (1).	1+1+1
16	Deflection occurs in one direction while entering (1); In the opposite direction when leaving (1); The field through the coil first increases and then decreases (1).	1+1+1
17	The needle aligns with the magnetic field (1); Without a field it points in no fixed direction (1); It is used to find direction by the earth's field (1).	1+1+1
	OR: Metal body connected to the earth through the earth wire (1); If the live wire touches the body, current passes to earth (1); Prevents shocks and fires (1).	1+1+1
18	(a) To take leakage current to earth (1); (b) The live wire carries the full current, and cutting it isolates the appliance (1); (c) Thick wire has low resistance and less heating (1).	1+1+1
	OR: The appliance is cut off from the circuit (1); The wire may still be live, so it is dangerous (1); Switch and fuse give protection (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Live wire is red and earth wire is green (1). (ii) $I = P/V = 2000/220 \approx 9.1$ A (1). (iii) The geyser draws about 9 A, more than the 5 A rating of the lighting circuit, which would overheat or blow the fuse (1). (iv) The fuse melts and breaks the circuit when the current is excessive (short circuit or overloading) (1); the earth wire carries leakage current from the metal body to the ground, avoiding shocks (1).	1+1+1+2
20	Depends on current (1); On the strength of the field (1); On the angle between wire and field (1); Reverse the current or field to reverse the force (1); Field up and current right (both along the page): force is out of the page, towards the observer (1).	1+1+1+1+1

	OR: While pushing in: the needle deflects one way (1); While held at rest: no deflection (1); While pulling out: deflection in the opposite direction (1); Induced current appears only when the field linked with the coil changes (1); Faster motion gives a larger deflection (1).	1+1+1+1+1

Mistakes to avoid:

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